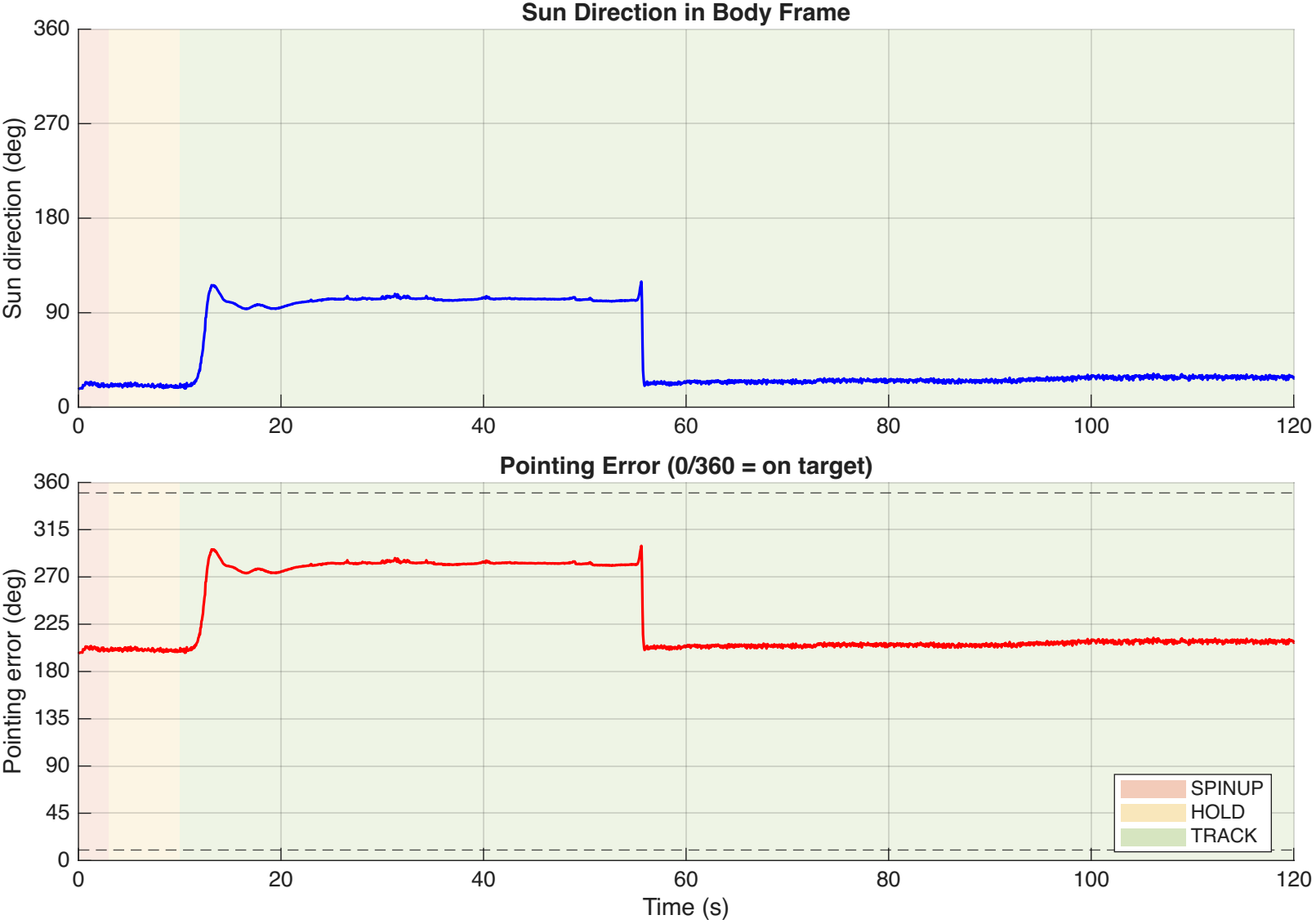
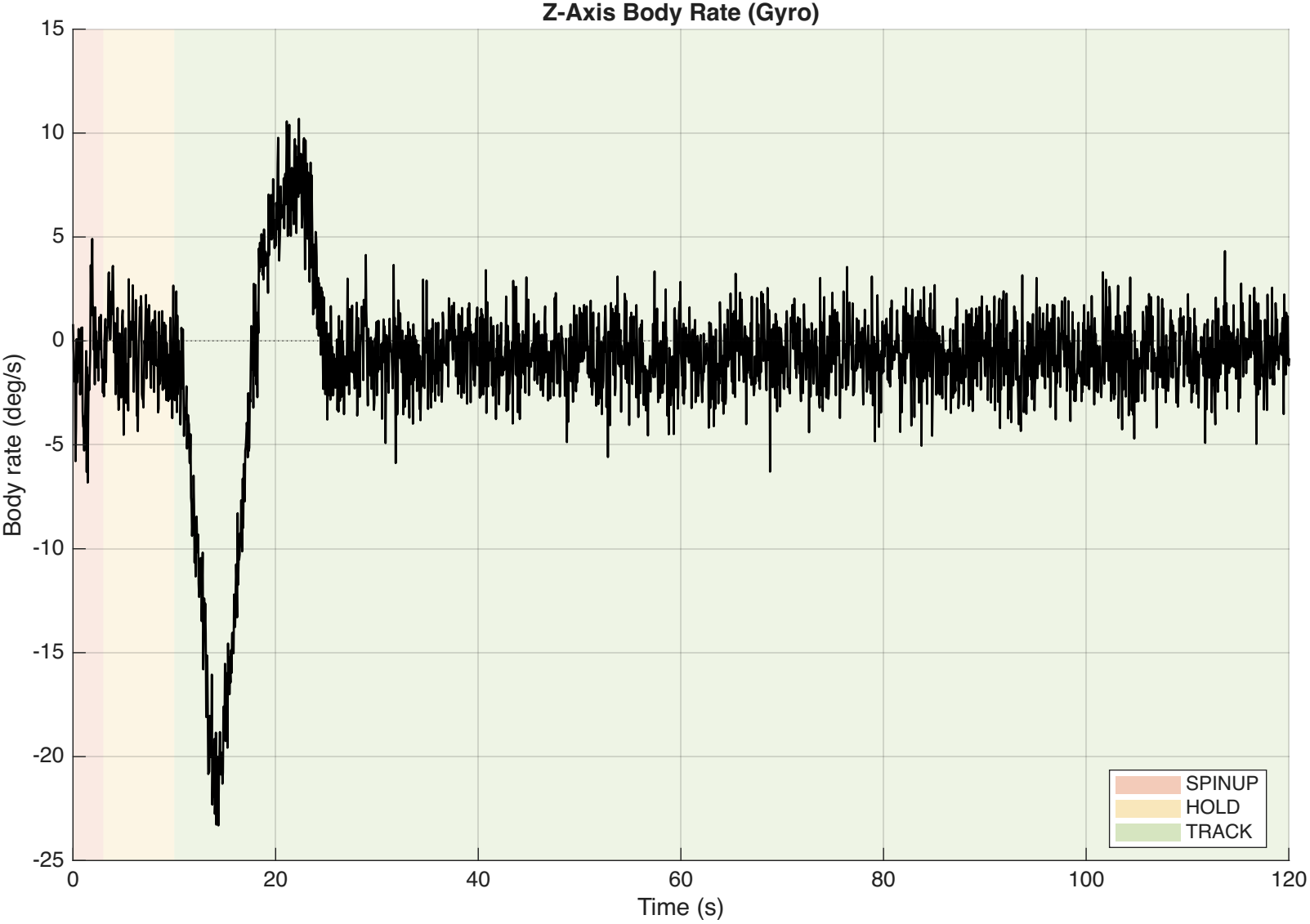
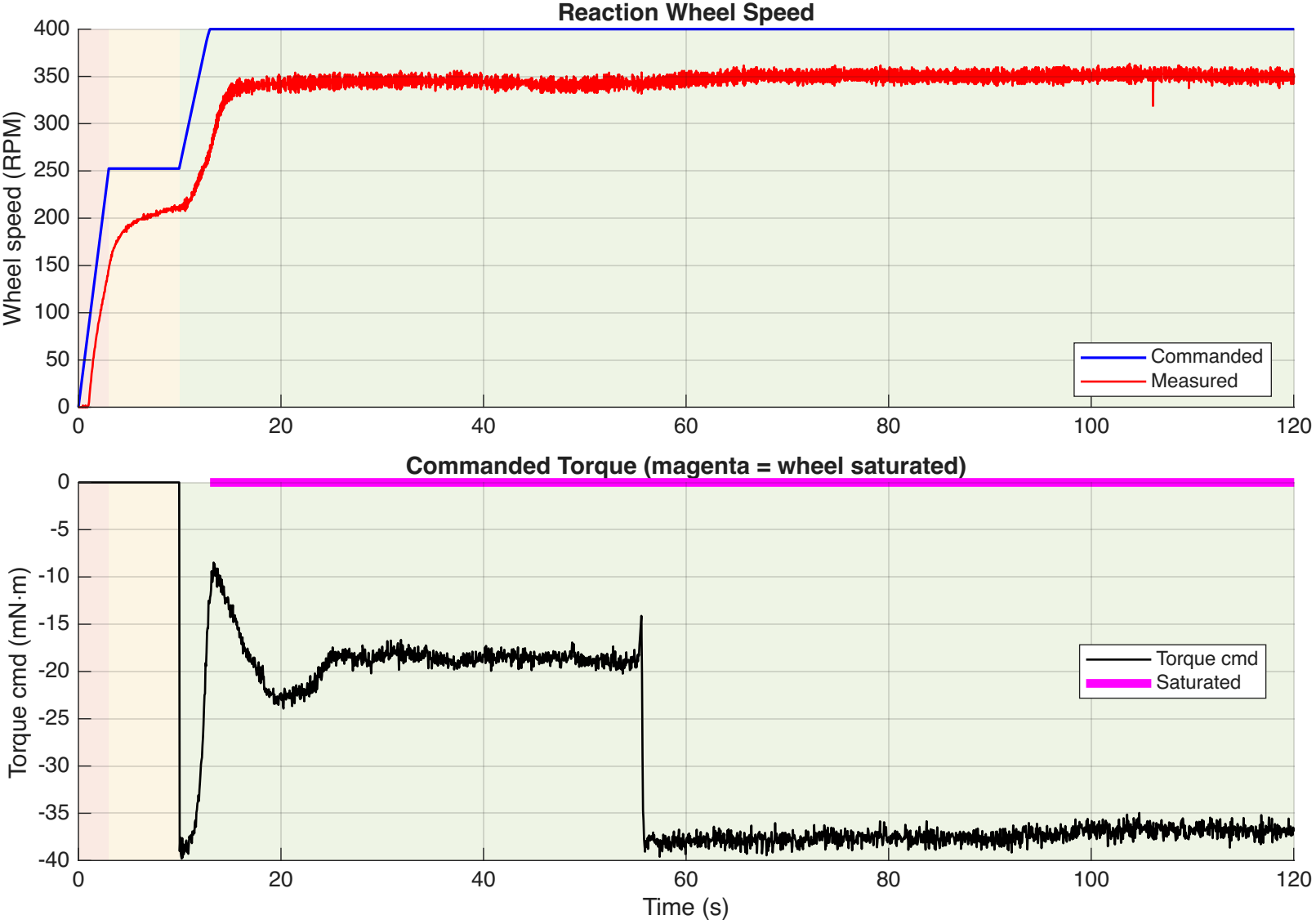
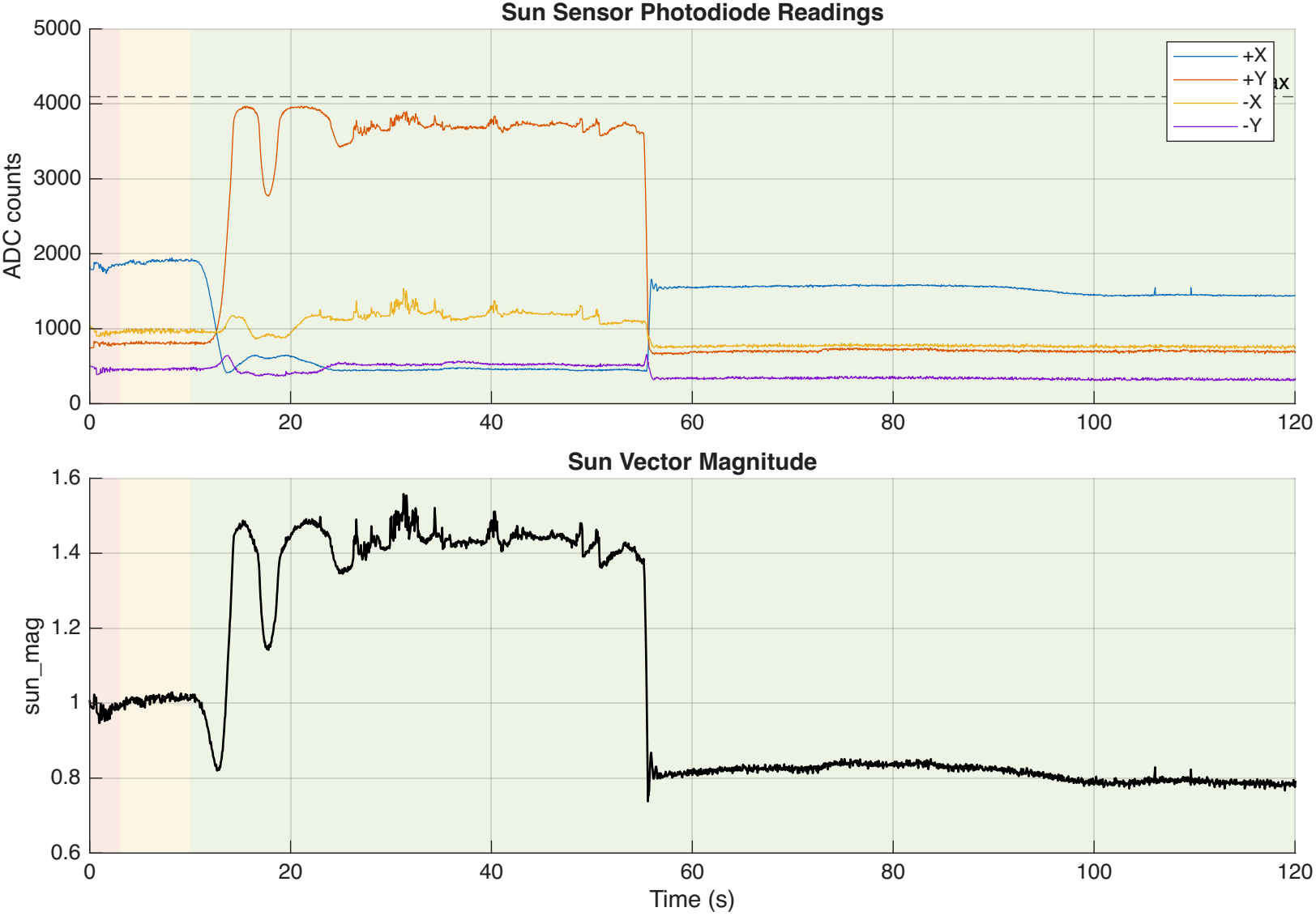
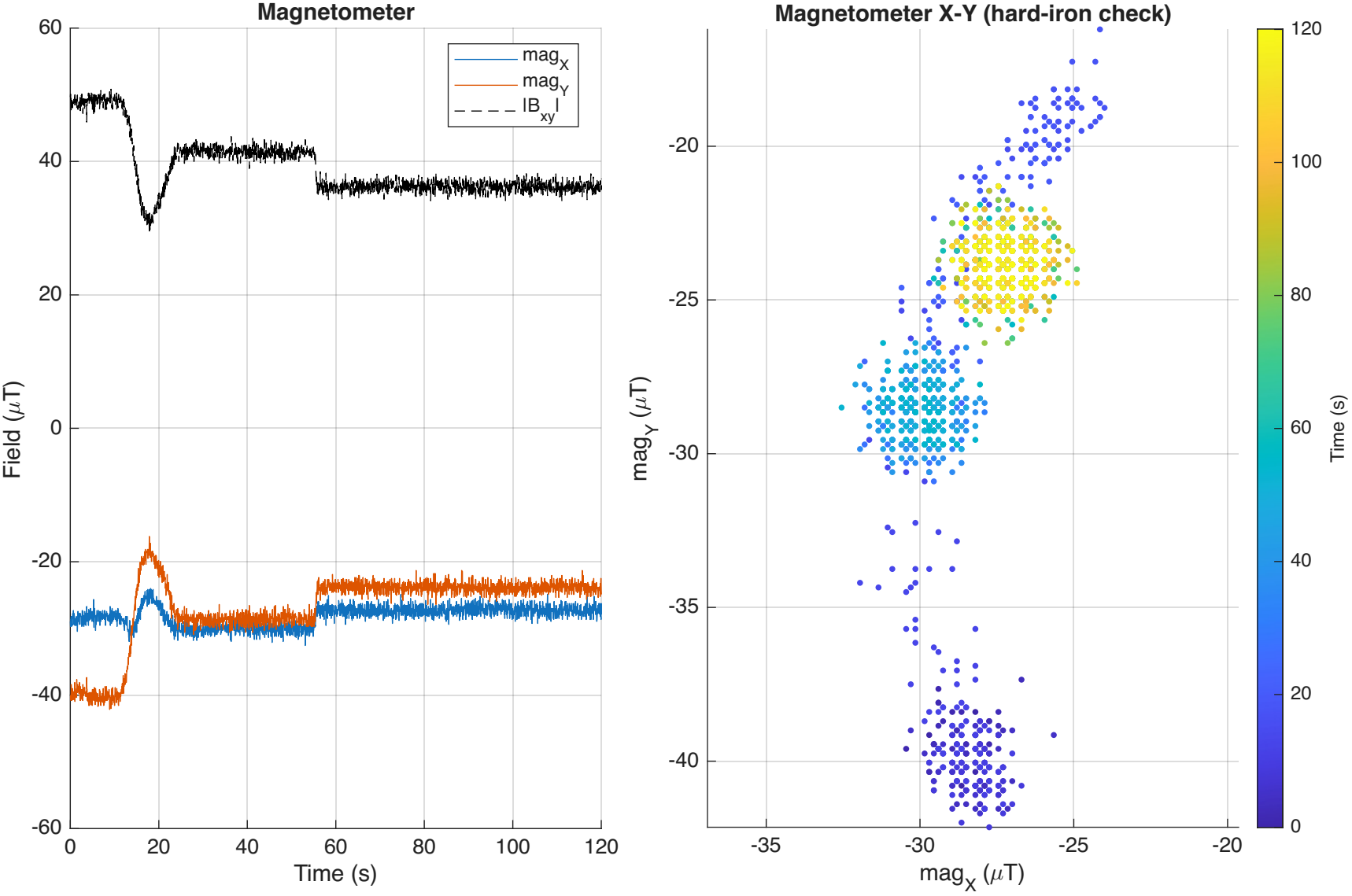
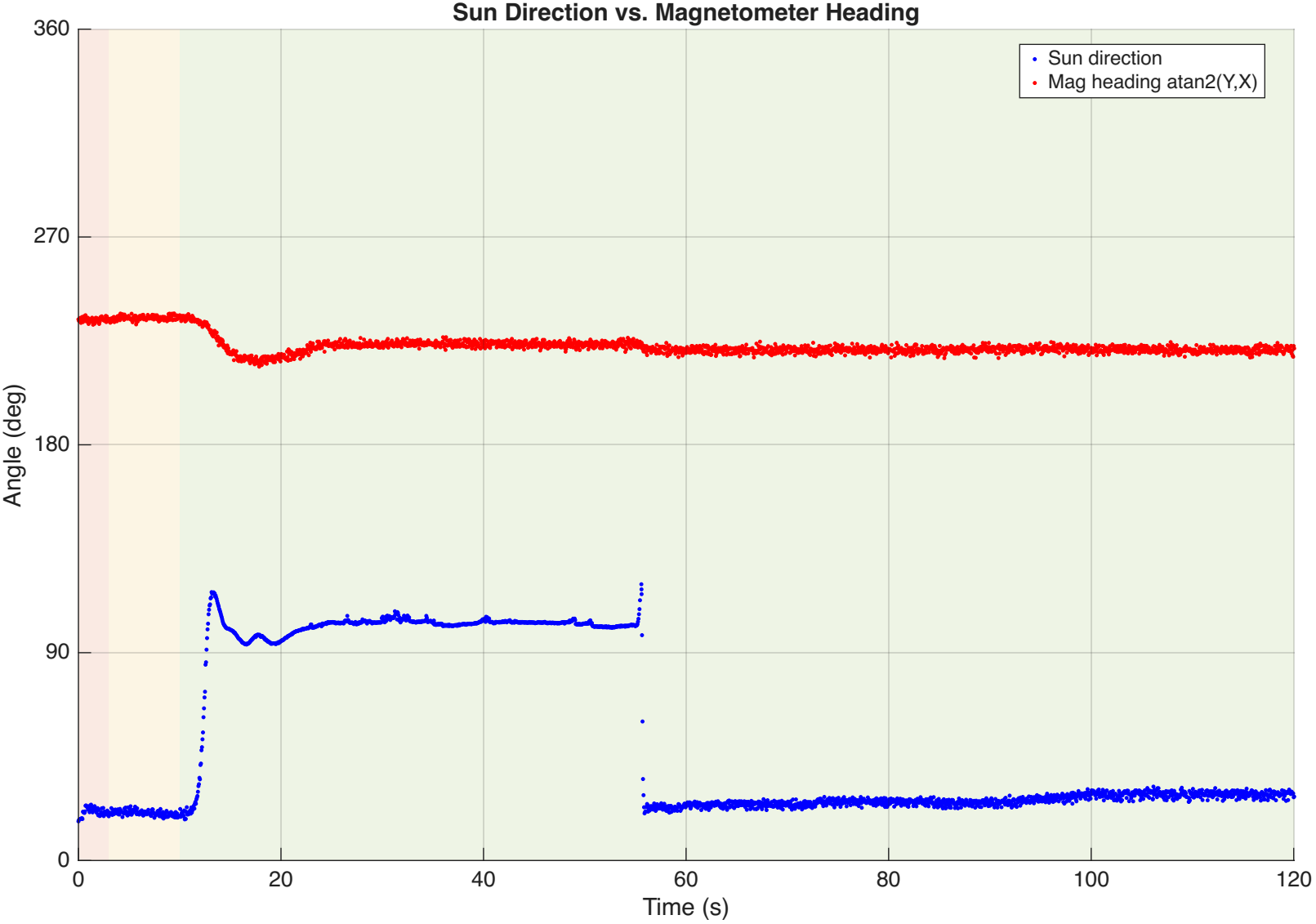

Sun Direction in Body Frame
Sun direction (deg)
360
270
180
90
0
0
20
40
60
80
100
120
Pointing Error (0/360 = on target)
Pointing error (deg)
360
315
270
225
180
135
90
45
0
SPINUP
HOLD
TRACK
Time (s)

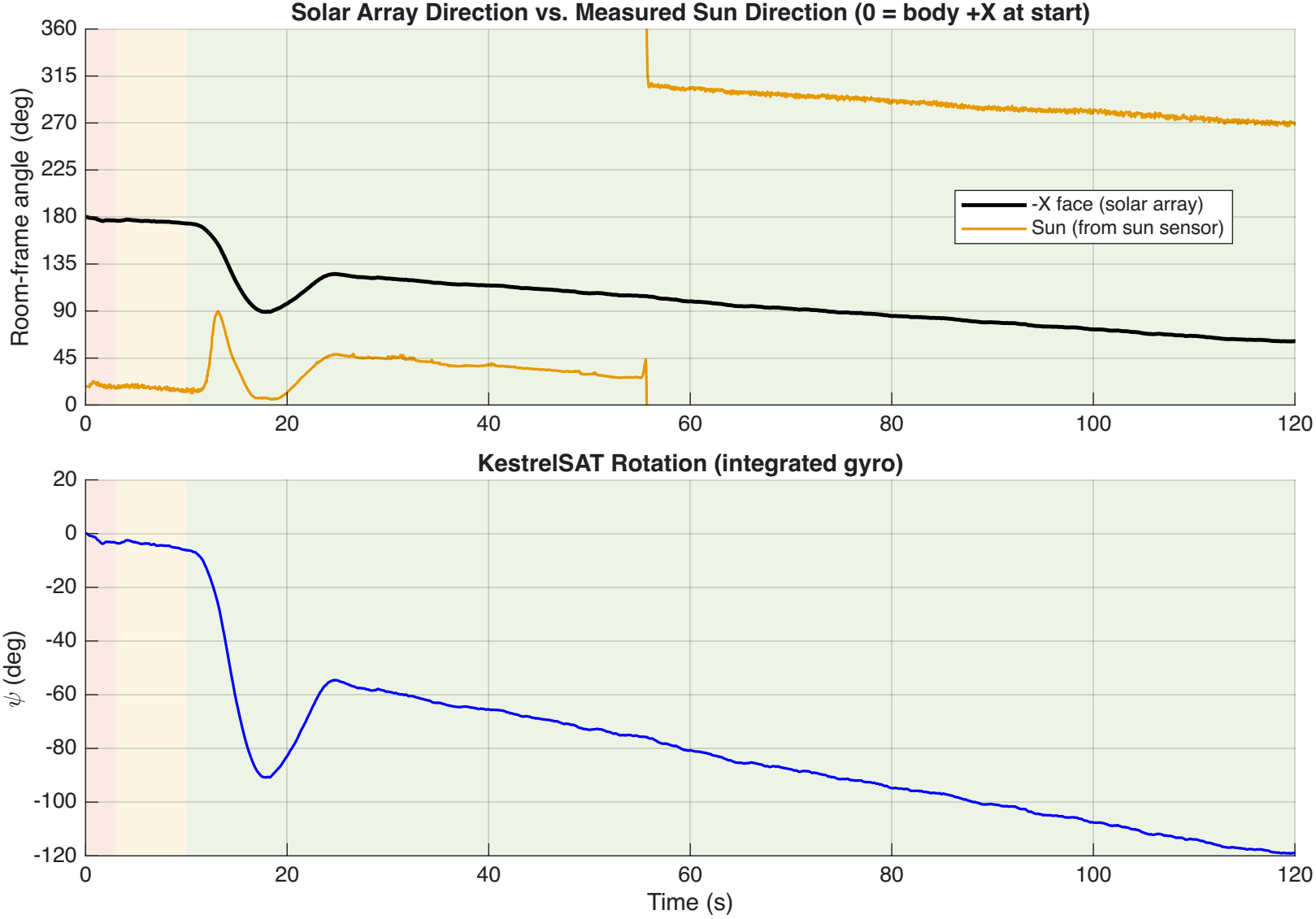

Z-Axis Body Rate (Gyro)
Body rate (deg/s)
Time (s)
15
10
5
0
-5
-10
-15
-20
-25
0
20
40
60
80
100
120
SPINUP
HOLD
TRACK

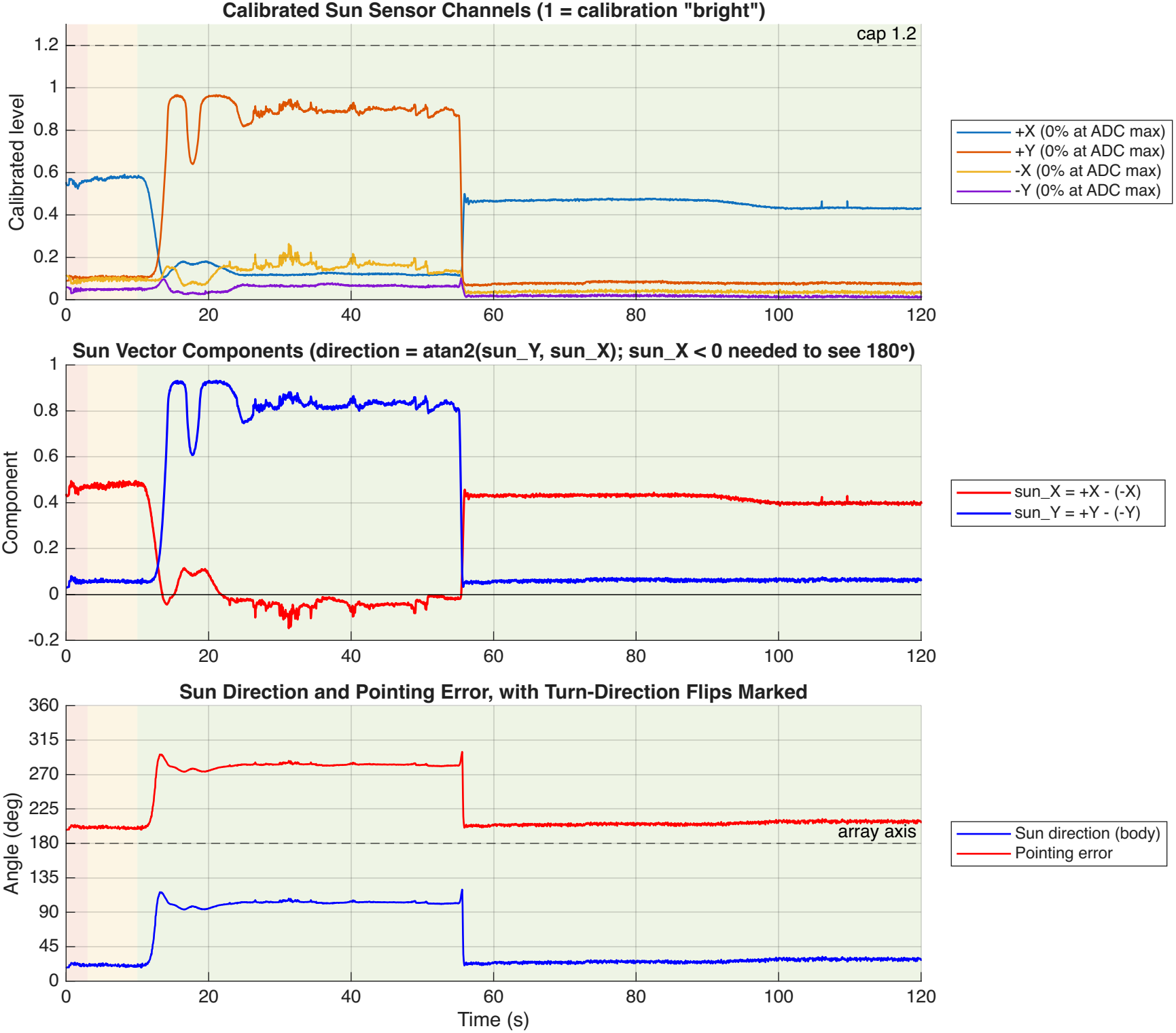

Reaction Wheel Speed
Wheel speed (RPM)
400
350
300
250
200
150
100
50
0
0
20
40
60
80
100
120
Commanded
Measured
Commanded Torque (magenta = wheel saturated)
Torque cmd (mN·m)
0
-5
-10
-15
-20
-25
-30
-35
-40
0
20
40
60
80
100
120
Time (s)
Torque cmd
Saturated

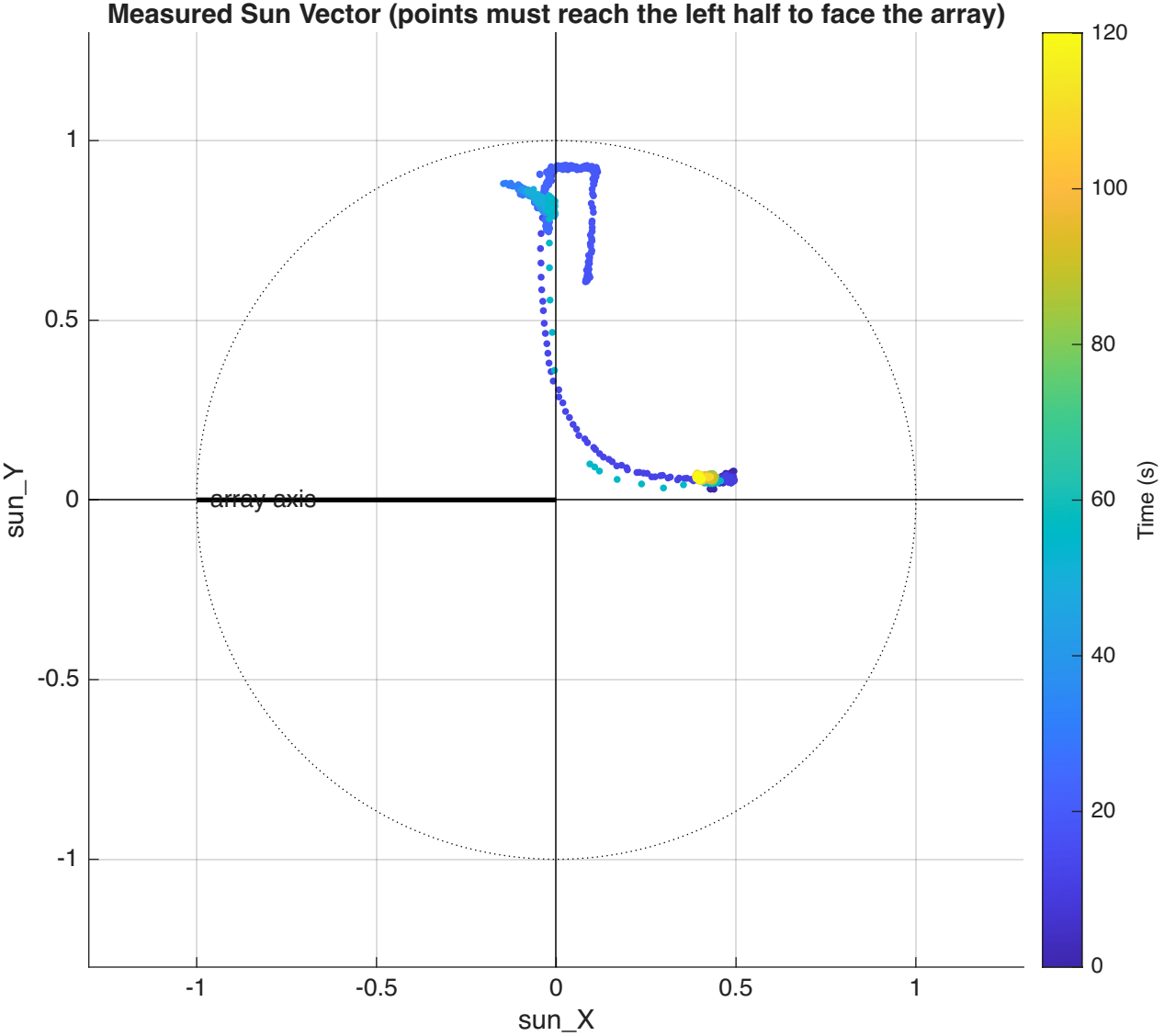

Sun Sensor Photodiode Readings
ADC counts
5000
4000
3000
2000
1000
0
0
20
40
60
80
100
120
+X
+Y
-X
-Y
Sun Vector Magnitude
sun_mag
1.6
1.4
1.2
1
0.8
0.6
Time (s)

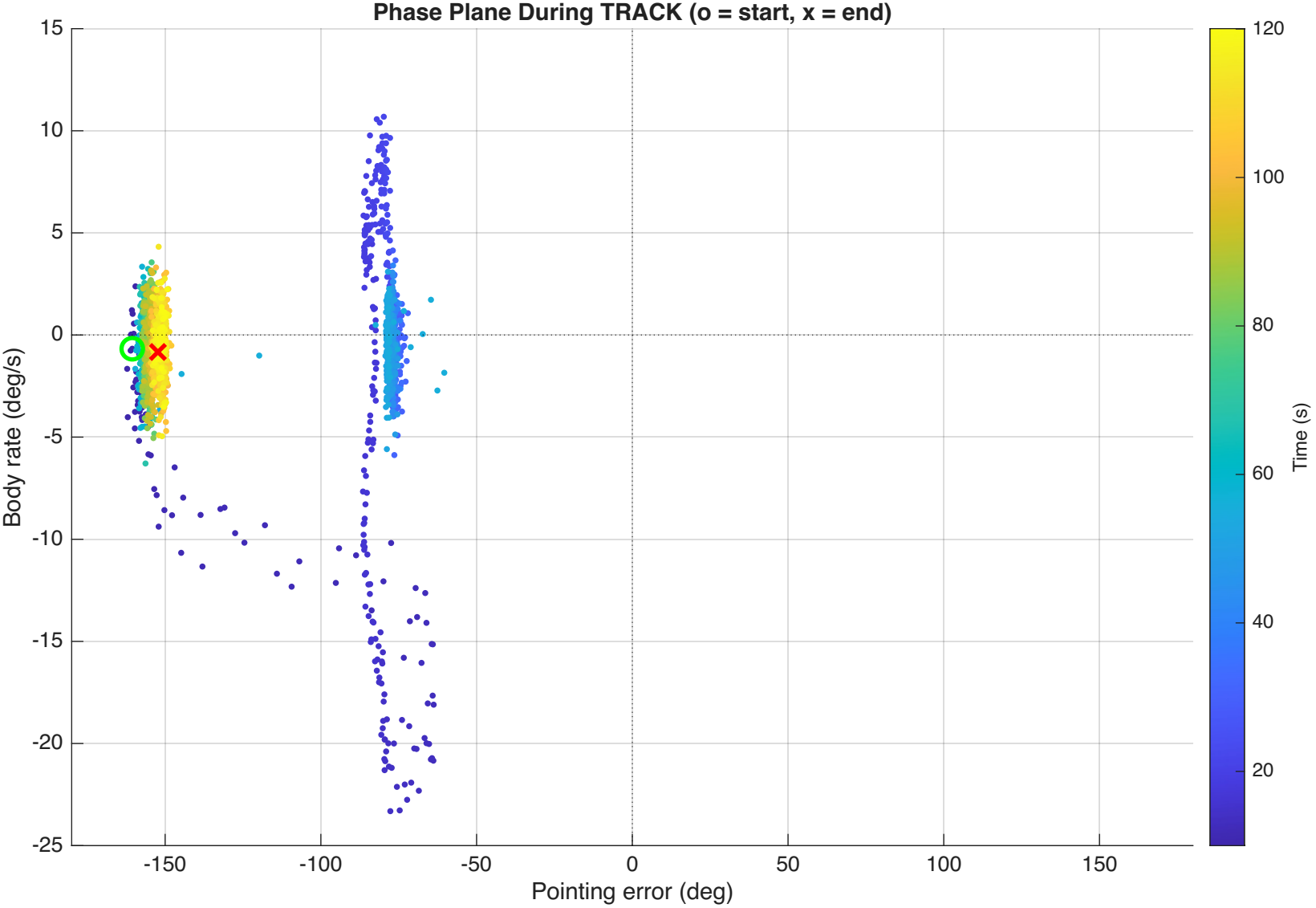

Magnetometer
mag_X
mag_Y
|B_xy|
Field (μT)
Time (s)
60
40
20
0
-20
-40
-60
0
20
40
60
80
100
120
Magnetometer X-Y (hard-iron check)
mag_Y (μT)
mag_X (μT)
-20
-25
-30
-35
-40
-35
-30
-25
-20
Time (s)
120
100
80
60
40
20
0

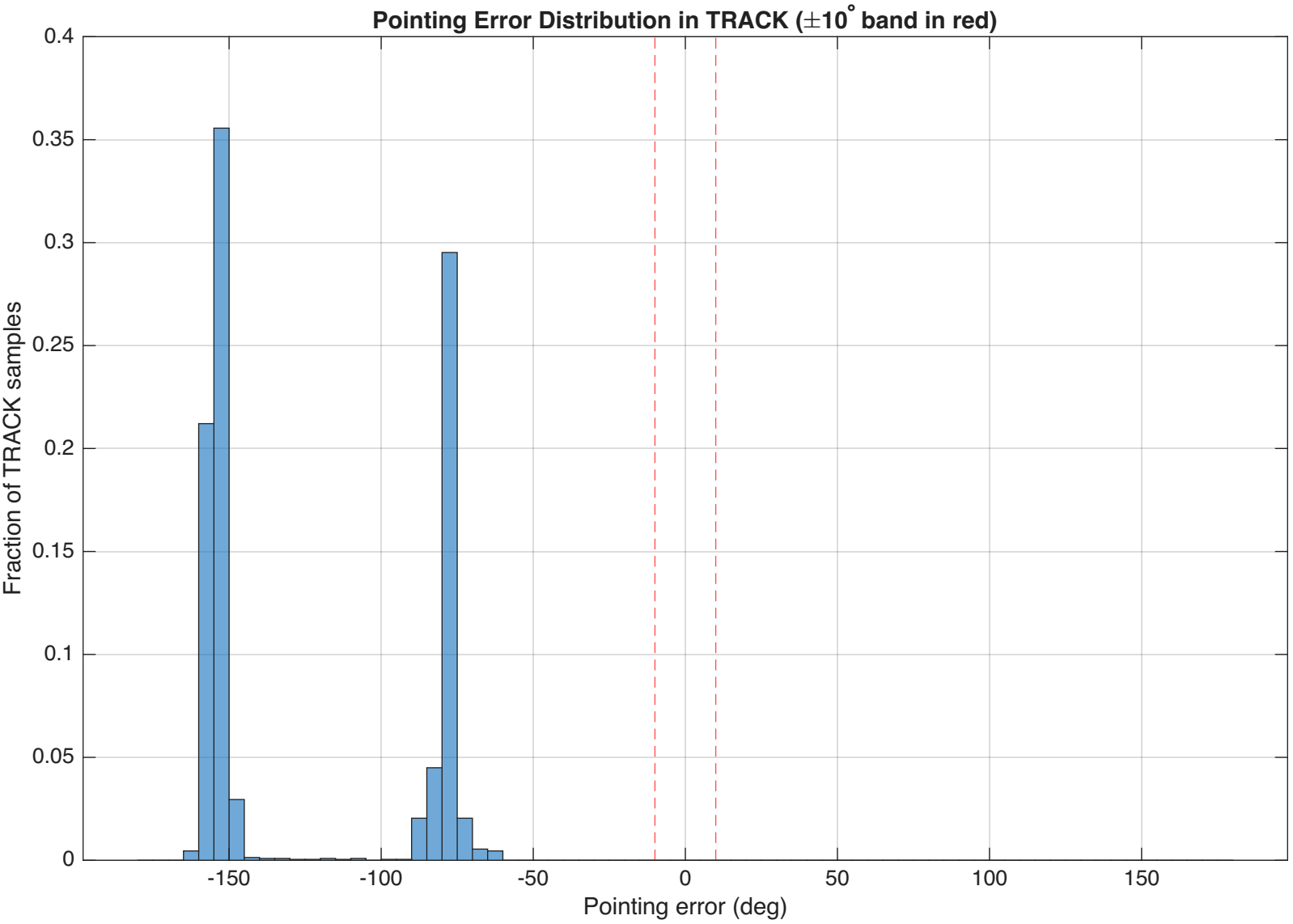

Sun Direction vs. Magnetometer Heading
Sun direction
Mag heading atan2(Y,X)
Angle (deg)
360
270
180
90
0
0
20
40
60
80
100
120
Time (s)


Solar Array Direction vs. Measured Sun Direction (0 = body +X at start)
Room-frame angle (deg)
360
315
270
225
180
135
90
45
0
0
20
40
60
80
100
120
-X face (solar array)
Sun (from sun sensor)
KestrelSAT Rotation (integrated gyro)
ψ (deg)
20
0
-20
-40
-60
-80
-100
-120
Time (s)


Calibrated Sun Sensor Channels (1 = calibration "bright")
cap 1.2
Calibrated level
+X (0% at ADC max)
+Y (0% at ADC max)
-X (0% at ADC max)
-Y (0% at ADC max)
Sun Vector Components (direction = atan2(sun_Y, sun_X); sun_X < 0 needed to see 180°)
Component
sun_X = +X - (-X)
sun_Y = +Y - (-Y)
Sun Direction and Pointing Error, with Turn-Direction Flips Marked
Angle (deg)
array axis
Sun direction (body)
Pointing error
Time (s)


Measured Sun Vector (points must reach the left half to face the array)
array axis
sun_Y
sun_X
Time (s)
1
0.5
0
-0.5
-1
-1
-0.5
0
0.5
1
120
100
80
60
40
20
0


Phase Plane During TRACK (o = start, x = end)
Body rate (deg/s)
Pointing error (deg)
Time (s)
15
10
5
0
-5
-10
-15
-20
-25
-150
-100
-50
0
50
100
150
120
100
80
60
40
20


Pointing Error Distribution in TRACK (±10° band in red)
Fraction of TRACK samples
Pointing error (deg)
0.4
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
-150
-100
-50
0
50
100
150